SAI SANTOSH CHEETHIRALA

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Engineer who builds measurement you can trust: fraud detection on payment graphs, revenue recovery reported against an untouched control arm, and causal attribution under distribution-free coverage guarantees. First-author paper submitting to TMLR.

EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability **Bengaluru | Jun 2026 – Jul 2026**

No Single Culprit — Causal Attribution Benchmark & Evaluation Harness for Multi-Hop RAG **github.com/santoshcheethiralame-dot/LINEUP**

Python, pytest, GitHub Actions (CI/CD), REST API, Streamlit, vLLM, BM25, PyTorch, HuggingFace Transformers, Conformal Prediction

- Built and shipped a 7.2K-LOC evaluation service — public REST API, pooled leaderboard, and Streamlit explorer — with 135 automated tests gating every change on GitHub Actions CI
- Replaced single-passage attribution with a calibrated set-valued method, lifting top-1 accuracy from 0.35 to 0.64 at 94% conformal coverage; first-author paper submitting to TMLR

PROJECTS

Untangle — Graph-Native Detection of Mule-Account Fraud Rings (UPI-style payments) **github.com/santoshcheethiralame-dot/UNTANGLE**

PyTorch Geometric, GraphSAGE, GAT, synthetic data generation, evaluation harness

- Modelled UPI-style payments as a single graph to detect coordinated mule-account rings, on the thesis that no individual hop in a laundering ring looks abnormal — which is the design of the crime and the reason per-transaction scoring cannot see it
- Built the synthetic generator adversarially against its own thesis: half of each ring is rented rather than fresh, so those mules are statistically identical to ordinary customers on every profile feature (account age log 6.82 vs 6.86), with tests asserting this parity holds — preventing a benchmark a decision tree could win
- Built and benchmarked GraphSAGE and GAT against tabular baselines on a purpose-built evaluation harness; ring extraction, API, and visualiser in progress

LIFT — Causal Revenue-Recovery Agent (Razorpay AI Buildathon, Track 03) **github.com/santoshcheethiralame-dot/LIFT**

TypeScript, causal inference, control-arm experiment design, 127 tests, zero runtime dependencies

- Built a revenue-recovery agent that reports what it causally caused rather than gross recovery, measured against a control arm left completely untouched, with the cost of acting charged against the same ledger
- Measured +24.4pp mean recovery lift across 30 independent batches (median +25.1pp, sd 4.6pp) and validated the estimator as near-unbiased against a true per-record counterfactual oracle (+24.4 reported vs +24.9 actual)
- Shipped with 127 tests running in ~15 seconds with no network, no API key, and zero runtime dependencies; reported the demo batch as running nearly a full standard deviation above the mean rather than quoting it as typical

tessera — Distributed, Sharded, Replicated Key/Value Store (Rust, std-only) **github.com/santoshcheethiralame-dot/TESSERA**

Rust (0 external crates), Raft, LSM-tree, Deterministic Simulation Testing, Linearizability Checking, MVCC

- Built a distributed, sharded, replicated key/value store in pure Rust with zero external crates: a from-scratch LSM storage engine (memtable, SSTables, write-ahead log, size-tiered compaction, MVCC) and per-shard Raft replication behind a coordinator and routing client (CockroachDB/TiKV shape, scaled down)

TECHNICAL SKILLS

Data & Analysis: Causal inference, control-arm experiment design, conformal prediction, statistical validation, benchmark design, evaluation harnesses

ML & Research: PyTorch, HuggingFace Transformers, conformal prediction, causal attribution, mechanistic interpretability (sparse autoencoders, activation & attribution patching), RAG, graph neural networks (GraphSAGE, GAT), 4-bit quantization

Languages: Python, C, Rust, TypeScript, JavaScript, SQL, WGSL

Systems & Backend: Distributed systems (Raft, LSM storage, MVCC), FastAPI, REST APIs, Docker, Linux, WebGPU/WGSL, vLLM

EDUCATION

PES University — B.Tech, Computer Science Engineering **Bengaluru | Sep 2024 – May 2028 (Expected)**

CGPA 7.95/10 (through Sem 4) — CNR Merit Scholarship (top 25% cohort), Sem 1 — Academic Distinction Award (SGPA 7.75+), Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, Linear Algebra, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University — Multimedia Domain Head **Oct 2025 – Mar 2026**

- Built a content pipeline and posting schedule for a 50+ member AI club, driving 5-6x growth in content views over three months
- National-Level Air Rifle Competitor - 10m Air Rifle, National School Games (a precision event built on sustained focus)